NUMERICAL OPTIMIZATION - BOOK CLUB

OPTIMISATION QUANT TEAM

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1. Fundamentals of Unonstrained Optimization

The concept of a local minimizer is introduced,

Definition (Weak local minimizer)**.** A point x^* is a *weak local minimizer* if there is a neighborhood [1] $\mathcal{N}$ of x^* such that $f(x^*) \leq f(x), \quad \forall x \in \mathcal{N}$.

On the other hand a *strict* local optimizer is where the inequality above is strict. A local minimizer x^* is called *isolated* if there exists a neighborhood in which x^* is the only local minimizer. Sufficient conditions for a point to be a local minimizer were presented as the following

Theorem 1 (Second-Order Sufficient Conditions)**.** Suppose that $\nabla^2 f$ is continous in an open neighborhood of x^* and that $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite. Then x^* is a strict local minimizer of f

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[1] A neighborhood of a point x, sometimes denoted $\mathcal{N}_x$ is an open set containing x. Defining which subsets of are considered open is the act of choosing a topology for your space. In this book they use the topology induced on $\mathbb{R}^n$ by the euclidean metric $(\|\cdot\|_2) : x \mapsto \sqrt{x^T x}$. For more on Euclidean topology, see here.

1.1. **Questions.**

i) *Do we use any of these?* Vplot has some unconstrained, least-squares algorithm.

ii) *What have you discussed previously on the topic of smooth optimization?* Interior point methods for optimizing SIMM delta.

iii) *How is the process of inverting a matrix affected by terrible condition number?* Which algorithms are used for matrix inversion? Is there some bound on iterations dependent on the condition number?

1.2. **Solutions.** The exercises chosen for this chapter are 2.1, 2.3, 2.10, 2.11 and 2.12.

1.2.1. *Exercise 2.1.* Compute i) the gradient and ii) the Hessian of the Rosenbrock [2] function

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

and iii) show that $x^* = (1, 1)$ is the only local minimizer.

i) The gradient of a function is the vector of all its partial derivatives

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}\right)^T.$$

Computing the two derivatives in x_1 and x_2 we get

$$\frac{\partial f}{\partial x_1} = -400(x_2 - x_1^2)x_1 - 2(1 - x_1)$$
$$\frac{\partial f}{\partial x_2} = 200(x_2 - x_1^2)$$

ii) The Hessian of a function is the matrix of all it's partial derivatives

$$\mathbf{H}_f = \left(\nabla^2 f\right)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

As a mnemonic, $\nabla = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_2}\right)$ and $\nabla^2 = \nabla\nabla^T$.

$$\frac{\partial^2 f}{\partial x_1} = -400\left(x_2 - 3x_1^3\right) + 2$$
$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = -400x_1$$
$$\frac{\partial^2 f}{\partial x_2} = 200$$

iii) For a local minimizer, $\nabla f = 0$ and $\nabla^2 f$ is positive definite. The first condition means to solve for which x_1, x_2 the equations

$$\frac{\partial f}{\partial x_1} = 0, \quad \frac{\partial f}{\partial x_2} = 0$$

hold. The second one is the easiest to start with, it implies the relation $x_2 = x_1^2$ for stationary points (a point for which $\nabla f(x) = 0$ is true) of f. Substituting into the former of the equations above we get

$$2(1 - x_1) = 0 \implies x_1 = 1$$

[2] There are higher dimensional generalizations of the Rosenbrock function which are classic problems to benchmark performance on n-dimensional, non-convex probelms.

which in turn means $x_2 = 1$. This means there is only one stationary point and thus, at most one local minimizer. Only if the Hessian is positive definite can the stationary point be a local minimizer. Let's check!

$$\nabla^2 f(1,1) = \begin{pmatrix} 802 & -400 \\ -400 & 200 \end{pmatrix}$$

A symmetric matrix is positive definite if all it's leading principal minors are positive[3]. The principal minors are the determinants of square sub-matrices starting from the upper left corner. The first one is the 1×1 matrix in the top left corner, i.e. $802 > 0$. The second one is

$$\det \nabla^2 f(1,1) = 802 \cdot 200 - 400^2 = 400 > 0$$

□

1.2.2. *Exercise 2.3.* Starting with f_1,

$$\frac{\partial f_1}{\partial x_i} = \frac{\partial a^T x}{\partial x_i} = \frac{\partial}{\partial x_i} \sum_i a_i x_i = a_i.$$

So $\nabla f_1 = a \in \mathbb{R}^n$. Since the gradient is constant and the Hessian consists of second derivatives, it is zero $\nabla^2 f_1 = \mathbf{0}$.

For f_2 we have

$$\begin{aligned} \frac{\partial f_2}{\partial x_k} &= \sum_{i,j} A_{ij} \frac{\partial (x_i x_j)}{\partial x_k} \\ &= \sum_{i,j} A_{ij} (\delta_{ik} x_j + x_i \delta_{jk}) \\ &= \sum_j A_{kj} x_j + \sum_i A_{ik} x_i \\ &= 2 A_k x \end{aligned}$$

where A_k is the kth column of A and δ_{ij} is the Kronecker delta[4]. The last inequality follows from A being symmetric. Thus

$$\nabla f_2 = (2A_1 x, \ldots, 2A_n x)^T = 2Ax.$$

For the Hessian, we need

$$\frac{\partial^2 f_2}{\partial x_l x_k} = \frac{\partial}{\partial x_l} \left(2 \sum_i A_{ik} x_i \right) = 2A_{lk}.$$

which gives

$$\nabla^2 f_2 = 2A$$

[3]S] Sylvester's criterion
[4]S] Kronecker delta

1.2.3. *Exercise 2.10.*

$$\frac{\partial \tilde{f}}{\partial z_i} = \sum_k \frac{\partial f}{\partial x_k} \frac{\partial x_k}{\partial z_i}$$

To begin with, we compute the second factor in the sum above

$$\frac{\partial x_k}{\partial z_i} = \frac{\partial}{\partial z_i} \left(\sum_j S_{kj} z_j + s_k \right) = S_{ki}.$$

Substituting this for $\frac{\partial x_k}{\partial z_i}$ in the first equation we get

$$\frac{\partial \tilde{f}}{\partial z_i} = \sum_k \frac{\partial f}{\partial x_k} S_{ki} = S_i^T \cdot \nabla f$$

with S_i being the ith column of S. This means[5]

$$\nabla \tilde{f}(z) = \begin{bmatrix} S_1 \nabla f \\ \vdots \\ S_n \nabla f \end{bmatrix} = S^T \nabla f(x)$$

For the Hessian, we need

$$\frac{\partial^2 f}{\partial z_i \partial z_j} = \sum_{k,l} \frac{\partial^2 f}{\partial x_k \partial x_l} \frac{\partial x_k}{\partial z_i} \frac{\partial x_l}{\partial z_j} = \sum_{k,l} \frac{\partial^2 f}{\partial x_k \partial x_l} S_{ki} S_{lj}$$

Note that we could have hade second derivatives of x w.r.t. z, but thankfully the first derivative is constant so we are off the hook! The expression above can be rewritten as:

$$\frac{\partial^2 f}{\partial z_i \partial z_j} = S_i^T \cdot \nabla^2 f \cdot S_j$$

So when assembling the whole Hessian matrix from the partial derivatives, we obtain

$$\nabla \tilde{f}(z) = S^T \nabla f(x) S$$

□

1.2.4. *Exercise 2.11.* Assuming a unit step size, the relation between subsequent steps is

$$x_{k+1} = x_k + p$$

We use this relation to make sure the affine relation between the iterates is preserved using this update rule. To begin with, the relation between the iterates z_k and x_k is

$$x_k = S z_k + s.$$

The step direction at each iteration is computed as

$$p_k = -B_k^{-1} \nabla f(x_k)$$

and

$$\tilde{p}_k = -\tilde{B}_k^{-1} \nabla \tilde{f}(z_k)$$

Given that the initial Hessians approximations are chosen s.t $\tilde{B}_0 = S^T B_0 S$ we can s] that

$$\tilde{p}_0 = -(S^T B_0 S)^{-1} S^T \nabla f(x_0) = -S^{-1} B_0^{-1} \nabla f(x_0)$$

[5] If S_i had b]n rows, we would have had $\nabla \tilde{f}(z) = S \nabla f(x)$

which, for z_{k+1}, means

$$\begin{aligned} z_{k+1} =& z_k + \tilde{p_k} = z_k - S^{-1}B_k^{-1}\nabla f(x_k) \\ =& (S^{-1}(x_k - s)) - S^{-1}B_k^{-1}\nabla f(x_k) \end{aligned}$$

Applying the affine transformation $v \mapsto Sv + s$ on both sides

$$\underbrace{x_k + B_k^{-1}\nabla f(x_k)}_{x_{k+1}} = Sz_{k+1} + s$$

So, given $\tilde{B}_k = S^T B_k S$, the above holds. Since we can choose the initial approximations such that this holds, it remains to verify that the rank one and BFGS update rules preserve this relationship. We begin with the rank-one update.

$$B_{k+1} \;=\; B_k \;+\; \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k}$$

where

$$s_k \;=\; x_{k+1} - x_k, \qquad y_k \;=\; \nabla f(x_{k+1}) - \nabla f(x_k).$$

We know that for $k = 0$, we have $x_k = Sz_k + s$ and $\tilde{B}_0 = S^T B_0 S$. If this update preserves the relation between the Hessian iterates, then the relation between the solution iterates will be preserved and continue to hold through further iterations. The quantities y_k and s_k relate to their˜counterparts as follows:

$$\tilde{s}_k \;=\; z_{k+1} - z_k = S^{-1}(x_{k+1} - x_k) = S^{-1}s_k, \qquad \tilde{y}_k \;=\; \nabla f(z_{k+1}) - \nabla f(z_k) = S^T y_k.$$

We now show the update satisfies behaves desirably by substituting in the above relations,

$$\begin{aligned} \tilde{B}_{k+1} &= \tilde{B}_k + \frac{(\tilde{y}_k - \tilde{B}_k\tilde{s}_k)(\tilde{y}_k - \tilde{B}_k\tilde{s}_k)^T}{(\tilde{y}_k - \tilde{B}_k\tilde{s}_k)^T\tilde{s}_k} \\ &= S^T B_k S + \frac{(S^T y_k - S^T B_k S S^{-1} s_k)(S^T y_k - S^T B_k S S^{-1} s_k)^T}{(S^T y_k - S^T B_k S S^{-1} s_k)^T S^{-1} s_k} \\ &= S^T B_k S + \frac{S^T (y_k - B_k s_k)(y_k - B_k s_k)^T S}{(y_k - B_k s_k)^T S S^{-1} s_k} \\ &= S^T \underbrace{\left(B_k + \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k} \right)}_{B_{k+1}} S \end{aligned}$$

The procedure for the BFGS update rule is identical. □

1.2.5. *Exercise 2.12.* This question is not entirely explicit in how to setup the problem, so I will make some assumptions to illustrate the effect of scaling. Given some "properly scaled" function $g(x_1, x_2)$ and take f to be the following function

$$f(x_1, x_2) = g(k_1 x_1, k_2 x_2)$$

where $k_1 \gg k_2$ are scale factors to make f poorly scaled. Given the Hessian has full rank, we can always choose a coordinate system in which the mixed partial derivatives are 0, so we will simply assume that is the case for $\frac{\partial^2}{\partial x_1 \partial x_2}$. This means the Hessian is:

$$\nabla^2 f = \begin{pmatrix} k_1^2 \frac{\partial^2 g}{\partial x_1^2} & 0 \\ 0 & k_2^2 \frac{\partial^2 g}{\partial x_2^2} \end{pmatrix}$$

The condition number of a matrix, $\kappa(A)$, is the quotient between its largest and smallest singular values (which for a square matrix is the absolute value of its eigenvalues). Denoting the set of eigenvalues of a matrix by Λ, one can write

$$\kappa(A) = \sup_{\lambda_1,\lambda_2\in\Lambda} \left|\frac{\lambda_1}{\lambda_2}\right|$$

In our case, that becomes:

$$\kappa\left(\nabla^2 f\right) = \frac{\left|k_1^2 \frac{\partial^2 g}{\partial x_1^2}\right|}{\left|k_2^2 \frac{\partial^2 g}{\partial x_2^2}\right|}$$

Now if we assume that "properly scaled" means that $|\frac{\partial^2 g}{\partial x_1^2}| \approx |\frac{\partial^2 g}{\partial x_2^2}|$ at the optimum then that means

$$\kappa\left(\nabla^2 f\right) \approx \frac{k_1^2}{k_2^2}$$

So if one axis has a characteristic length of $\sim 10^6$ times larger than the other, the condition number will be 10^{12}!

2. Theory of Constrained Optimization

The formulation of constrained problems in this book,

$$\min_{x\in\mathbb{R}^n} f(x) \qquad \text{subject to} \quad \begin{cases} c_i(x) & = 0, \quad i \in \mathcal{E}, \\ c_i(x) & \geq 0, \quad i \in \mathcal{I} \end{cases} \tag{1}$$

is standard in the literature, except for the sign chosen of the inequality constraint functions ($i \in \mathcal{I}$) which may vary from author to author. By defining the feasible set as

$$\Omega \equiv \{x \in \mathbb{R}^n | c_i(x) = 0, \forall i \in \mathcal{E}, c_i(x) \geq 0, \forall i \in \mathcal{I}\} \tag{2}$$

the authors present a very elegant formulation of the constrained problem as

$$\min_{x\in\Omega} f(x).$$

2.1. Farkas' lemma.

In the proof of the first order optimality conditions (Theorem 12.1 in the book) the authors use

Lemma 1 (Farkas)**.** Let the cone K be defined as

$$K = \{By + Cw | y \geq 0\}.$$

Given any vector $g \in \mathbb{R}^n$ we have either that $g \in K$ (i.e. $g = By + Cw$ for some $y \geq 0, w \in \mathbb{R}^n$) or that there exists $d \in \mathbb{R}^n$ satisfying

$$g^T d < 0, \quad B^T d \geq 0, \quad C^T d = 0 \tag{3}$$

I felt the rest of the proof, after Farkas' lemma was a bit thin so I thought I would expand a bit here. The proof assumes the constraint qualification LICQ, that the gradients of the active constraints $\nabla c_i(x^*)$, $i \in \mathcal{A}(x^*)$ at the optimum are linearly independent. Then they show that there exists a vector λ^* of Lagrange multipliers which satisfies the KKT conditions. They have previously showed that for all vectors d in the tangent cone at the optimum $T_\Omega(x^*)$, the following holds

$$d^T \nabla f(x^*) \geq 0.$$

They also show that if the LICQ hold, $T_\Omega(x^*) = \mathcal{F}(x^*)$. Recalling the definition of $\mathcal{F}(x^*)$, the set of first order feasible directions,

$$\mathcal{F}(x^*) = \left\{ d \;\middle|\; \begin{array}{ll} d^T \nabla c_i(x^*) \geq 0, & \forall i \in A_{\mathcal{I}}(x^*) \cap \mathcal{I} \\ d^T \nabla c_i(x^*) = 0, & \forall i \in \mathcal{E} \end{array} \right\}$$

By identifying B with the inequality constraint gradients and C with the equality constraint gradients in Farkas' lemma, we see that there must be some vector $\lambda_i^* \geq 0, i \in \mathcal{A}(x^*) \cap \mathcal{I}$ for which

$$\nabla f(x^*) = \nabla c_i(x) \lambda^*$$

since all vectors d that satisfy the latter two of the conditions in (3) do not satisfy $d^T \nabla f(x^*) < 0$. The proof is then finished by verifying that this λ^*, together with the local solution x^* does in fact satisfy all the KKT conditions.

2.2. **Solutions.** The exercises chosen for this chapter are 12.1, 12.2, 12.4, 12.4, 12.11 (not **b)**) and 12.22.

2.2.1. *Exercise 12.4.* Just as in the proof of Theorem 2.5[6] we prove by contradiction. Assume x is a local but not global solution. Then there must exist some z for which $f(z) < f(x)$. From the convexity of the feasible region, we can produce any convex combination of x and z and rest assured of it's feasibility. So, if we consider the line between x and z, which must be contained in Ω by convexity, we have from the convexity of f

$$f(\lambda z + (1-\lambda)x) \leq \lambda f(z) + (1-\lambda) f(x) < f(x)$$

Thus, in an arbitrarily tiny neighborhood (radius λ) around x, there must exist points with a lower function value. This contradicts the assumption that x is a local solution and it must thus hold that any solution which is a local solution must also be a global. □

2.2.2. *Exercise 12.11.*

a) The feasible set is $\Omega = \{x \in \mathbb{R}^2 \mid x_2 \geq 0, x_2 \leq x_1^2\}$. Let's define the constraint functions as $c_1(x) = x_2 \geq 0$ and $c_2(x) = x_1^2 - x_2 \geq 0$. The tangent cone $T_\Omega(x^*)$ is the set of all directions d for which there exists a sequence of feasible points $\{z_k\} \subset \Omega$, $z_k \to x^*$ and a sequence of positive scalars $\{t_k\}$, $t_k \to 0^+$ as $k \to \infty$ such that:

$$d = \lim_{k\to\infty} \frac{z_k - x^*}{t_k}$$

Now, let $d = (d_1, d_2)$ be a vector in $T_\Omega(x^*)$, and let $z_k = (x_{1k}, x_{2k})$. We can then write the above as:

$$(d_1, d_2) = \lim_{k\to\infty} \left(\frac{x_{1k}}{t_k}, \frac{x_{2k}}{t_k} \right)$$

Since each $z_k \in \Omega$, it must satisfy the constraints. We use the constraints to furnish restrictions on the possible values of d_1 and d_2. Starting out with c_1, we divide it by $t_k > 0$ to get the following condition:

$$\frac{x_{2k}}{t_k} \geq 0 \overset{k\to\infty}{\Longrightarrow} d_2 \geq 0$$

[6]Found on page 16 of the book.

Dividing c_2 by t_k as well, we get

$$\frac{x_{2k}}{t_k} \leq \frac{x_{1k}^2}{t_k}.$$

where can rewrite the right-hand side as:

$$\frac{x_{2k}}{t_k} \leq \left(\frac{x_{1k}}{t_k}\right) \cdot x_{1k}.$$

Now, take the limit as $k \to \infty$. We know $\lim \frac{x_{2k}}{t_k} = d_2$, $\lim \frac{x_{1k}}{t_k} = d_1$, and $\lim x_{1k} = 0$ (since $z_k \to x^*$).

$$d_2 \leq \left(\lim_{k\to\infty} \frac{x_{1k}}{t_k}\right) \cdot \left(\lim_{k\to\infty} x_{1k}\right)$$
$$d_2 \leq d_1 \cdot 0$$
$$d_2 \leq 0$$

Combining $d_2 \geq 0$ and $d_2 \leq 0$, we must have $d_2 = 0$. There is no restriction on d_1. Thus, finally, the tangent cone is

$$T_\Omega(x^*) = \{(d_1, 0) | d_1 \in \mathbb{R}\}.$$

In the above, we have assumed that the tangent cone is not empty, in part **d)**, we show that is the case.

How about the set of first-order feasible directions, $\mathcal{F}$? The gradients of the constraint functions, $c_1(x) = x_2$, $c_2(x) = x_1^2 - x_2$, are

$$\nabla c_1(x^*) = (0, 1)$$
$$\nabla c_2(x^*) = (0, -1).$$

The set of first-order feasible solutions is the set of directions d (we only have inequality constriants in this case) for which the following conditions holds

$$d^T \nabla c_i(x) \geq 0, \quad \forall i \in \mathcal{A}(x) \cap \mathcal{I}$$

In order for $d^T \nabla c_i(x^*) \geq 0$ for both $i = 0$ and $i = 1$ we must have $d = (a, 0)^T$. Thus, $\mathcal{F}(x^*) = T_\Omega(x^*)$.

b) **skip until next session** LICQ are not satisfied since the gradients of the constraint functions are not linearly independent at $x^* = (0, 0)^T$

$$\nabla c_1(x^*) = (0, 1)$$
$$\nabla c_2(x^*) = (0, -1)$$

with $c_1(x) = x_2$, $c_2(x) = x_1^2 - x_2$.

c) If the objective function is $f(x) = -x_2$ the Lagrangian is

$$\begin{aligned}\mathcal{L}(x, \lambda) = -x_2 - \sum_{i\in\{1,2\}} \lambda_i c_i(x) &= -x_2 - \lambda_1 x_2 - \lambda_2(x_1^2 - x_2) \\ &= -\lambda_2 x_1^2 + (\lambda_2 - \lambda_1 - 1)x_2\end{aligned}$$

So, (12.34a) becomes:

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = \begin{pmatrix} 0 \\ \lambda_2 - \lambda_1 - 1 \end{pmatrix} \stackrel{!}{=} 0$$

Choosing Lagrange multiplies such that $\lambda_2 - \lambda_1 = 1$ satisfies the above condition. Since the LICQ does not hold, λ^* is not unique. For (12.34b)

and (12.34c) we already know x^* is feasible. To satisfy (12.34d) we can pick $\lambda_1 = 1$, $\lambda_2 = 2$, for example. Lastly, (12.34e) is satisfied since both constraint functions are zero at x^*.

d)

$$z_k = \begin{pmatrix} t_k \\ t_k^2 \end{pmatrix}$$

with $t_k = 1/k$

2.2.3. *Exercise 12.22.* We begin by making explicit the implications of the properties of $c_i(x), i \in \mathcal{E}$ being linear and $-c_i(x), i \in \mathcal{I}$ being convex. The defining property of a linear function is the following

$$f(ax + by) = af(x) + bf(y)$$

and for a function to be convex means

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y), \qquad \lambda \in [0,1].$$

In words, the function between two points x, y is bounded above by the straight line connecting $f(x), f(y)$. Additionally, the negative of a convex function is *concave*, which has the same definition, but with $\leq$ switched for $\geq$.

Now for a set to be convex, we require any convex combination of points within the set to also be members of the set[7]. That is,

$$\lambda x + (1-\lambda)y = z \overset{!}{\in} \Omega$$

To show this holds, we need to show z is feasible.

$$c_i(z) = 0, \quad i \in \mathcal{E}$$
$$c_j(z) \geq 0, \quad i \in \mathcal{I}$$

Starting with the equality constraints

$$0 \overset{!}{=} c_i(z) = c_i(\lambda x + (1-\lambda)y) = \lambda c_i(x) + (1-\lambda)c_i(y) = 0$$

where the second inequality follows from linearity of $c_i, i \in \mathcal{E}$ and the third from $x, y \in \Omega$. For the inequality constraints we have

$$0 \overset{!}{\leq} c_i(z) = c_i(\lambda x + (1-\lambda)y) \geq \lambda c_i(x) + (1-\lambda)c_i(y) \geq 0$$

where the second inequality comes from $c_i, i \in \mathcal{I}$ being concave. □

[7]A convex combination is a linear combination with non-negative coefficents summing to one. If you allow the coefficents to take any value, but still require them to sum to one, it is called an *affine* combination.